

Exercise 3: Concentration Bounds for Matrices

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Problem 1 (Covariance estimation):

Let Σ be a population covariance matrix, i.e. $\Sigma = \mathbb{E}\mathbf{x}\mathbf{x}^\top$, $\mathbf{x} \in \mathbb{R}^d$, and Σ_n be the approximate covariance matrix $\Sigma_n = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$ constructed from a given set of samples $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$.

We consider the case where $\mathbf{x}_i, \mathbf{x}$ are normally distributed, i.e. $\mathbf{x}_i, \mathbf{x} \sim \mathcal{N}(0, \sigma^2)$. Our goal is to estimate the distance $\|\Sigma_n - \Sigma\|_{\text{op}}$.

For this exercise, we will need the following variant of Bernstein's inequality:

Theorem 1. Let $\{X_i\}_{i=1}^n$ be independent sub-exponential random variables. Then

$$\Pr\left(\left|\sum_{i=1}^n (X_i - \mathbb{E}[X_i])\right| \geq t\right) \leq 2 \exp\left(-c \cdot \min\left(\frac{t^2}{\sigma^2}, \frac{t}{K}\right)\right),$$

where $\sigma^2 = \sum_{i=1}^n \|X_i\|_{\Psi_1}^2$ and $K = \max_i \|X_i\|_{\Psi_1}$.

Note that since the random variables are i.i.d., we can further simplify the result using $\sigma^2 = nK^2$ and rewrite the result (using $t = \delta n$) as

$$\Pr\left(\left|\sum_{i=1}^n (X_i - \mathbb{E}[X_i])\right| \geq \delta\right) \leq 2 \exp\left(-cn \cdot \min\left(\frac{\delta^2}{K^2}, \frac{\delta}{K}\right)\right).$$

Prove the following statements:

1. Show that one can choose an appropriate $\epsilon \in [0, \frac{1}{2}]$ for an ϵ -net over the sphere $\mathcal{S}^{d-1}$, such that

$$\|\Sigma_n - \Sigma\| \leq 2 \max_{\mathbf{v} \in \mathcal{N}} \left| \frac{1}{n} \sum_{i=1}^n \langle \mathbf{x}_i, \mathbf{v} \rangle^2 - \mathbb{E} \langle \mathbf{x}, \mathbf{v} \rangle^2 \right|.$$

Hint: For $\mathbf{x} \in \mathbb{R}^n$ and $\mathcal{N}$ an ϵ -net of the sphere $\mathcal{S}^{n-1}$, it holds that

$$\sup_{\mathbf{y} \in \mathcal{N}} \langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\|_2 \leq \frac{1}{1 - \epsilon} \sup_{\mathbf{y} \in \mathcal{N}} \langle \mathbf{x}, \mathbf{y} \rangle.$$

2. Assume that $\|\Sigma\| = 1$. Fix a vector $\mathbf{v} \in \mathcal{S}^{d-1}$ and let $\mathbf{x} \in \mathcal{N}(0, \sigma^2)$. Show that

$$\Pr(\langle \mathbf{x}, \mathbf{v} \rangle^2 \geq s) \leq 2 \exp(-s),$$

i.e. $\langle \mathbf{x}, \mathbf{v} \rangle^2$ is a sub-exponential random variable.

3. Using the last result combined with Theorem 1, show that

$$\Pr\left(\max_{\mathbf{v} \in \mathcal{N}} |S(\mathbf{v})| \geq \delta\right) \leq 2 \exp(-c_2 n),$$

for an appropriate constant $c_2 > 0$.

4. What do you conclude about the distance $\|\Sigma_n - \Sigma\|$?

Problem 2 (Matrix calculus):

In this problem, $\mathbf{X}$ and $\mathbf{Y}$ denote $d \times d$ symmetric real matrices.

1. Prove that $\|\mathbf{X}\| \leq t$ if and only if $-t\mathbf{I} \preceq \mathbf{X} \preceq t\mathbf{I}$, where $\preceq$ denotes the Loewner order.
2. Let $\mathbf{X}, \mathbf{Y}$ be commuting matrices, both being PSD. Prove that $\mathbf{X} \preceq \mathbf{Y}$ implies $\mathbf{X}^2 \preceq \mathbf{Y}^2$.
3. Show by example that the previous property may fail for general symmetric real matrices.